

ArtData™ Module

AD-AC — ArtData™ Agent Communication Module

ArtData™ Standard System

Module ID: AD-AC

Standard System: ArtData™

Category: AI & Data Integrity Standards (AI)

Subcategory: Agent Communication Integrity

Version: 1.0

Status: Canonical · Module

Compatibility: ArtData™ Standard · MTVF™ · AI Governance Frameworks

Canonical Language: English

About the Module

What the Agent Communication Module Is

The **ArtData™ Agent Communication Module (AD-AC)** defines structural transparency conditions for messages exchanged between AI systems.

As agents increasingly interact with other agents, services, and automated decision systems, communication integrity becomes critical.

Without structural controls, systems cannot reliably determine message origin, sender identity, message integrity, or decision responsibility.

AD-AC sets minimum requirements so AI-generated messages stay traceable and accountable.

What This Module Changes

In many AI environments automated messages are exchanged without clear sender identity.

This creates risks such as:

- automated misinformation propagation
- loss of accountability
- unreliable decision chains
- compromised system trust

The Agent Communication Module establishes a structural rule:

every AI-to-AI message must contain a verifiable origin reference.

Canonical Definition

ArtData™ Agent Communication is a structured integrity framework ensuring that messages exchanged between automated systems include identifiable origin, message integrity reference, and responsible entity declaration.

Communication integrity ensures that AI systems remain traceable and accountable within automated decision environments.

Scope

The Agent Communication Module applies to automated communication between:

- AI agents
- machine learning services
- automated decision systems
- API-based AI services
- autonomous system networks

The module focuses on structural traceability of automated messages.

Structural Requirements

To satisfy ArtData™ Agent Communication conditions, the following must be established.

Sender Identification

Each automated message must identify the sending system.

Minimum requirement:

- system identifier
 - service reference or agent ID
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Message Integrity Reference

Messages must include an integrity verification reference.

Examples:

- message hash
 - digital signature
 - verification token
-

Timestamp

Each automated message must include a time reference.

Minimum requirement:

- message timestamp
 - system time reference
-

Responsible Entity Declaration

A responsible organization must be identifiable.

Minimum requirement:

- organization name
 - system operator reference
-

Minimum Implementation Framework (MIF)

Implementation Steps

Step 1 — Assign Agent Identifier

Each AI system must possess a unique identifier.

Examples:

- agent ID
 - system reference
 - service identifier
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Step 2 — Attach Message Timestamp

Every automated message must include a timestamp.

This allows communication events to be reconstructed.

Step 3 — Apply Message Integrity Reference

Attach a verification reference to each message.

Examples:

- message hash
 - digital signature
 - verification token
-

Step 4 — Declare Responsible Entity

The organization operating the AI system must be identifiable.

Minimum information:

- organization name
 - system operator reference
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Architecture Position

The Agent Communication Module represents the communication integrity layer within the ArtData™ architecture.

ARTDATA™ DATA INTEGRITY STRUCTURE

Dataset Origin (AD-P)
↓
Dataset Integrity (AD-I)
↓
AI Training Dataset (AD-AI)
↓
Synthetic Data Transparency (AD-S)
↓
Agent Communication Integrity (AD-AC)
↓
Responsible Entity

Communication integrity ensures that automated systems remain traceable within multi-agent environments.

Use Case 1

Autonomous AI Agent Networks

A company deploys multiple AI agents coordinating logistics operations.

Without communication integrity controls, it is difficult to determine which agent initiated a decision.

Using the Agent Communication Module:

- each agent has a unique identifier
- every message contains a timestamp
- communication integrity references are attached

Result:

- automated decision chains become traceable
 - system debugging becomes possible
 - operational accountability increases
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Use Case 2

AI Service API Communication

An AI service communicates with external AI platforms through APIs.

Without communication integrity, messages may be modified or misattributed.

Using the Agent Communication Module:

- sender system identity is attached
- message integrity references are applied
- responsibility is traceable

Result:

- AI communication becomes verifiable
 - service interactions become more reliable
 - trust between automated systems increases
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Canonical Closing Statement

As artificial intelligence systems begin communicating with each other, message integrity becomes essential.

The ArtData™ Agent Communication Module ensures that automated communication remains identifiable, verifiable, and accountable.

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Canonical Source

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